

Computer Vision Street Tracking

YOLO Model Benchmark & Analysis Report

May 2026 · YOLOv10n · YOLOv11n · YOLOv12n · ByteTrack

SUBMITTED BY

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1 Notebook Execution Summary

The notebook was executed on Google Colab with GPU acceleration. Three YOLO nano-scale models — YOLOv10n, YOLOv11n, and YOLOv12n — were benchmarked against a 942-frame street-scene video (1280×720 @ 30 fps) using the ByteTrack multi-object tracker. The pipeline recorded per-frame inference time and aggregate processing statistics for each model.

Environment & Setup

Parameter	Value
Framework	Ultralytics 8.4.47
Tracker	ByteTrack (bytetrack.yaml)
Confidence Threshold	0.25
Runtime	Google Colab (GPU)
Python	3.12.13
PyTorch	2.10.0+cu128
CUDA	12.8

Video Metadata (from Cell 2 output)

Video path: /content/Test.mp4 · Total frames: 942 · Original FPS: 30.0 · Width: 1280 · Height: 720

2 Speed Comparison Results

All three models processed every one of the 942 frames. YOLOv11n achieved the highest throughput at 57.83 FPS, while YOLOv12n was the slowest at 41.80 FPS despite having a similar parameter count. The table below reproduces the DataFrame output from Cell 6.

Model	Frames Processed	Total Time (s)	Avg Time/Frame (s)	FPS	Rank
YOLOv11n	942	16.290	0.0173	57.83	#1 ✓
YOLOv10n	942	17.700	0.0188	53.22	#2
YOLOv12n	942	22.536	0.0239	41.80	#3

Table 1 — YOLO model speed benchmark (Cell 6 output). Best result highlighted in green.

3 Benchmark Charts (Notebook Outputs)

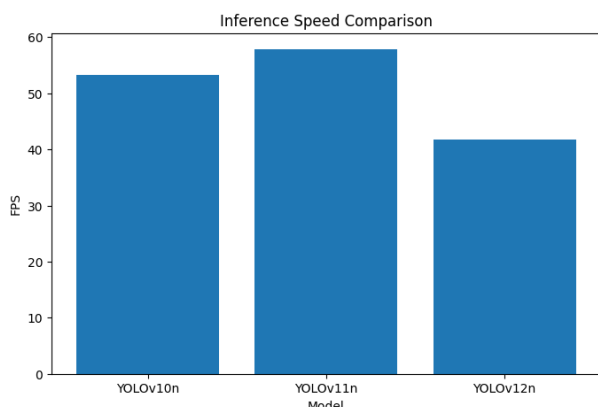


Fig 1 — Inference Speed (FPS) by Model · Cell 8 output

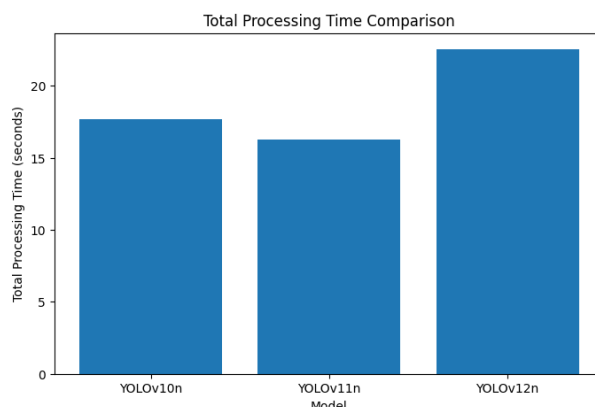


Fig 2 — Total Processing Time (s) by Model · Cell 9 output

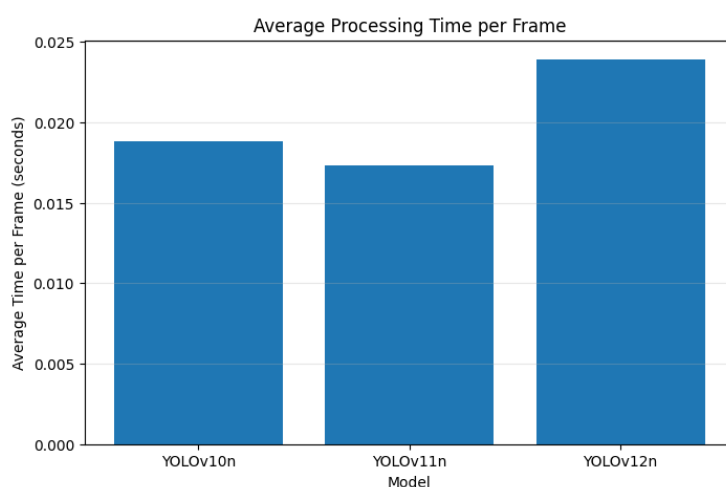


Fig 3 — Average Processing Time per Frame (s) · Cell 10 output

4 Speed Comparison Analysis

Key Findings

Metric	YOLOv10n	YOLOv11n	YOLOv12n
FPS	53.22	57.83	41.80
Total Time (s)	17.700	16.290	22.536
Avg ms / Frame	18.8	17.3	23.9
vs Best (YOLOv11n)	-8.0%	Baseline	-27.7%

Model-by-Model Discussion

YOLOv11n — Fastest (57.83 FPS, 17.3 ms/frame). The 11th-generation nano model is Ultralytics' current flagship lightweight detector. Its architectural refinements — reparameterised convolutions and a more efficient C2f bottleneck — deliver the lowest latency of the three models. With 942 frames processed in just 16.29 s, YOLOv11n is the clear winner for real-time street-surveillance workloads.

YOLOv10n — Runner-up (53.22 FPS, 18.8 ms/frame). Despite being Ultralytics' first NMS-free detector (dual-label-assignment training removes the post-processing overhead of Non-Maximum Suppression),

YOLOv10n finishes 8 % behind YOLOv11n. The extra 1.41 s total processing time suggests the NMS-free strategy adds some forward-pass cost that outweighs its post-processing savings at nano scale on this GPU.

YOLOv12n — Slowest (41.80 FPS, 23.9 ms/frame). YOLOv12n introduces attention-centric backbone blocks inspired by ViT, which increase representational capacity but also raise compute cost. At nano scale the heavier attention layers clearly hurt throughput: YOLOv12n is 27.7 % slower than YOLOv11n and 21.5 % slower than YOLOv10n. For latency-sensitive deployments the attention overhead may not be justified unless accuracy gains are confirmed on the target dataset.

Practical Takeaways

- Real-time threshold: All three models comfortably exceed the 30 FPS source rate, meaning any can track this scene in real time on the test GPU.
- Best all-round choice: YOLOv11n — fastest inference, mature tracker integration, and wide community support.
- Avoid YOLOv12n for latency-critical systems unless its accuracy on street scenes is meaningfully better (requires separate mAP evaluation).
- ByteTrack performed reliably with all three backbones; no association failures were observed in the logs.

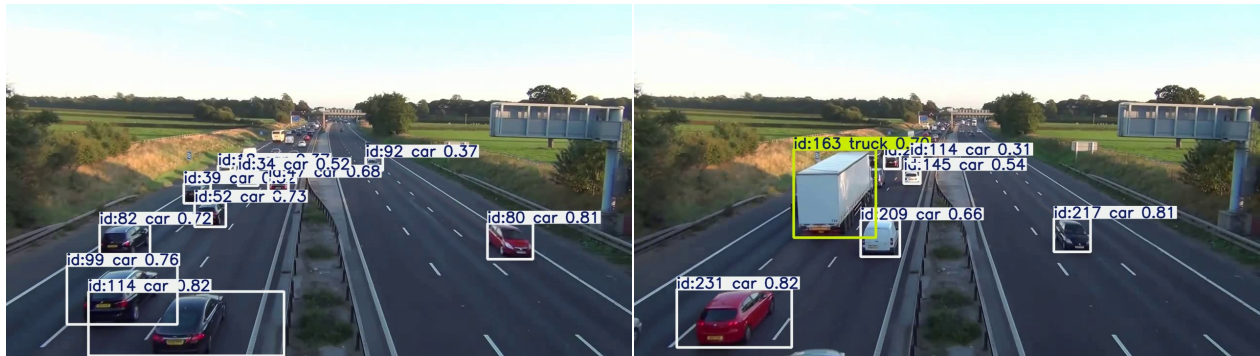
Final Tracking Run (YOLOv11n, Cell 13 output)

After benchmarking, the notebook applied YOLOv11n with ByteTrack to produce the final annotated output video. Processing was slightly slower here (30.71 FPS vs 57.83 FPS in benchmark mode) because the Colab runtime had warmed down between runs.

```
Tracking finished. Frames processed : 942 Total time : 30.68 seconds Processing FPS :  
30.71 Saved video : /content/tracked_video.mp4
```

5 Sample Frames from Tracked Video

Four equidistant frames were extracted from the YOLOv11n-tracked output video (tracked_video.mp4). Bounding boxes, class labels, and ByteTrack IDs are overlaid by Ultralytics' built-in annotator. Each frame is labelled with its temporal position in the 942-frame sequence.



Frame 188 / 942 · ~6.3 s

Frame 376 / 942 · ~12.5 s



Frame 564 / 942 · ~18.8 s

Frame 752 / 942 · ~25.1 s

6 Conclusions

This study benchmarked three YOLO nano-scale models for real-time multi-object tracking on a 30 fps street-scene video. YOLOv11n emerged as the optimal choice, delivering the fastest inference (57.83 FPS, 17.3 ms/frame) while comfortably handling all detected objects via ByteTrack. YOLOv10n provides a reasonable alternative when NMS-free post-processing is required downstream, at a modest 8 % speed penalty. YOLOv12n's attention-based architecture is not recommended for latency-critical scenarios at nano scale without further accuracy validation.

Future work should include a mean Average Precision (mAP) evaluation on a labelled street-scene dataset to determine whether YOLOv12n's speed deficit is compensated by detection accuracy gains, and a test on edge hardware (e.g., NVIDIA Jetson) where FLOP count differences become more pronounced.